Scene Characteristic Mining-Based Semisupervised Network for Ship Detection in SAR Images

Yuang Du
The National Laboratory of Radar Signal Processing
Xidian University
Xi'an, China
yuangd631@163.com

Lan Du
The National Laboratory of Radar Signal Processing
Xidian University
Xi'an, China
dulan@mail.xidian.edu.cn

Yuchen Guo
The Academy of Advanced Interdisciplinary Research
Xidian University
Xi'an, China
ychguo@xidian.edu.cn

Yu Shi
The National Laboratory of Radar Signal Processing
Xidian University
Xi'an, China
shiyu1213@126.com

Zilin Wang
The National Laboratory of Radar Signal Processing
Xidian University
Xi'an, China
wzlwzl12345@qq.com

2023 IEEE International Radar Conference (RADAR) | 978-1-6654-8278-3/23/$31.00 ©2023 IEEE | DOI: 10.1109/RADAR54928.2023.10371054

***Abstract*—Deep synthetic aperture radar (SAR) ship detection networks can achieve excellent performance by training with lots of target-level labels. However, obtaining target-level labels is extremely difficult in practice. In this paper, a scene characteristic mining-based semi-supervised network (SCMS-Net) is proposed for SAR ship detection to address this issue. This network can leverage scene-level labels of SAR images for enhanced performance when the target-level labels are limited. Our approach extends the fully supervised RefineDet network by incorporating a dedicated scene characteristic mining branch (SCMB). By effectively utilizing scene-level labels, SCMB can extract scene characteristics in SAR images that are beneficial for representing ships and clutter, thereby achieving higher performance. Additionally, a hierarchical test process is proposed in this paper. It can reduce false alarms in inland and inshore scenes, as well as missing ships in the offshore scene. Experiments based on an authoritative SAR ship detection dataset validate the effectiveness of our method.**

Keywords—semi-supervised learning, ship detection, scene characteristic mining, hierarchical test process, synthetic aperture radar (SAR)

I. Introduction

Synthetic Aperture Radar (SAR) can acquire a large range of surface information and is less affected by illumination and weather constraints. SAR ship detection is a crucial aspect of SAR technology, playing a significant role in ensuring maritime security and monitoring marine traffic. The Constant False Alarm Rate (CFAR) algorithm [1], as the most widely studied and classic method for SAR ship detection, has attracted significant research interest from scholars over the past several decades. However, the CFAR methods exhibit limited performance in some complex scenes, e.g., inland and inshore scenes [2].

In recent years, many researchers have extended and applied the convolutional neural network (CNN) [3-5] to SAR ship detection. Du et al. [2] combined an attention module with RefineDet [3] to alleviate the clutter interference in SAR images, thus achieving superior performance in complex scenes. An et al. proposed DrBox-v2 [6], a one-stage detector that utilizes focal loss and a hard negative mining mechanism to improve detection performance. Due to their remarkable feature extraction capabilities, the CNN-based detectors have emerged as the dominant approach in the SAR ship detection field.

Fig. 1. unlabeled SAR images

Fig. 2. SAR images with only scene-level labels

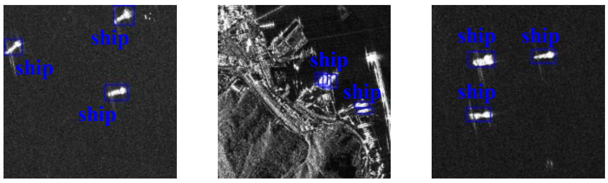


Fig. 3. SAR images with only target-level labels

However, training the above detectors requires a significant number of images with target-level labels. In practice, obtaining the target-level labels of SAR images is extremely difficult. To reduce reliance on target-level labels, researchers have given significant attention to semi-supervised learning [7-8]. During the training process of semi-supervised detectors, besides utilizing a limited number of images with target-level labels, a substantial number of unlabeled or scene-level labeled images can also be leveraged. From Fig. 1, Fig. 2 and Fig. 3, it can be observed that labeling the scene type of SAR images is much easier compared to labeling the exact positions of ships. Therefore, we focus on studying semi-supervised detection methods.

Some researchers have studied semi-supervised detection methods. Xu et al. [7] proposed a box jittering mechanism, which can select unlabeled images with reliable pseudo boxes for network training. Sohn et al. [8] proposed STAC, a method that leverages a large amount of unlabeled images in a self-

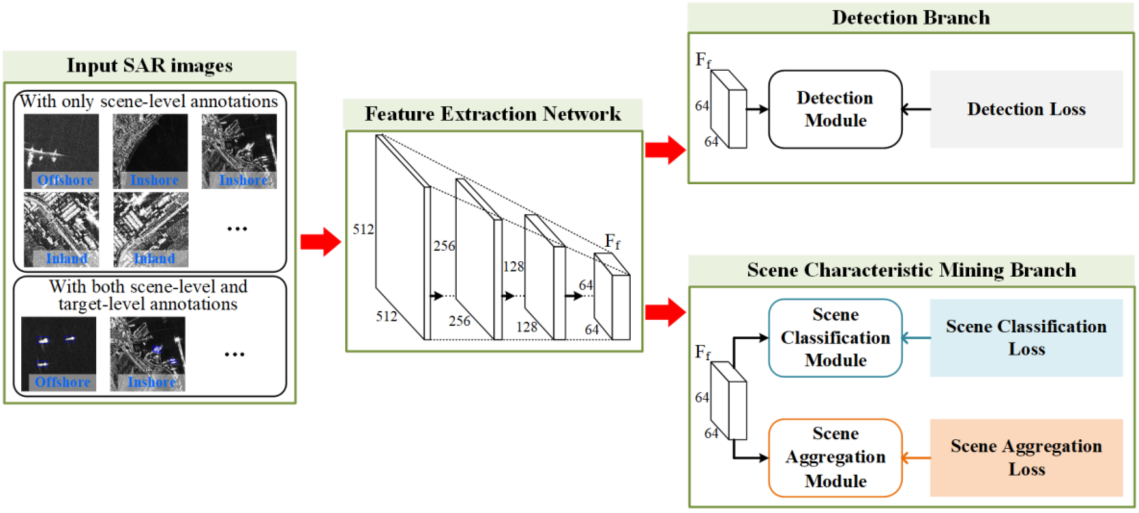


Fig. 4. The overall framework of SCMS-Net

learning manner. Different from these methods, we propose a novel scene characteristic mining-based semi-supervised network (SCMS-Net) for SAR ship detection. The SCMS-Net is capable of utilizing a substantial quantity of SAR images with solely scene-level labels and a limited number of SAR images with both target-level and scene-level labels at the same time for training. Specifically, a dedicated scene characteristic mining branch (SCMB) is proposed to leverage scene-level labels. In the SCMB, a scene classification loss and a scene aggregation loss are designed to learn the characteristics of each scene. As depicted in Fig. 2, SAR images from distinct scenes exhibit diverse characteristics regarding the likelihood of ship presence and the type of clutter. And by learning these scene characteristics, the feature representation capability of the network for ships and clutter will be enhanced, which contributes to achieving higher performance.

Besides, we design a hierarchical test process. Once the scene category of SAR images is recognized in the test process, we will employ specialized detection strategies that are specifically designed for each scene. During the design of different detection strategies, we take into full consideration the characteristics of different scenes. This test process can facilitate the reduction of false alarms in inland and inshore scenes, while also reducing missing ships in offshore scenes.

II. NETWORK STRUCTURE OF SCMS-NET

A. Overall Framework

We utilize RefineDet [2] as the baseline, it is a typical single-stage detector with good detection performance for multi-scale ship targets. And compared with other CNN-based detectors, RefineDet has a good trade-off in speed and accuracy. The overall framework of SCMS-Net is illustrated in Fig. 4. The SCMS-Net comprises four key components, and each component plays a significant role in the overall functioning of the network:

1) Input SAR images: The input SAR images are categorized into two types: SAR images solely labeled with scene-level labels and SAR images labeled with both target-level and scene-level labels. During training, the network simultaneously receives these two types of SAR images with different annotation levels as input.

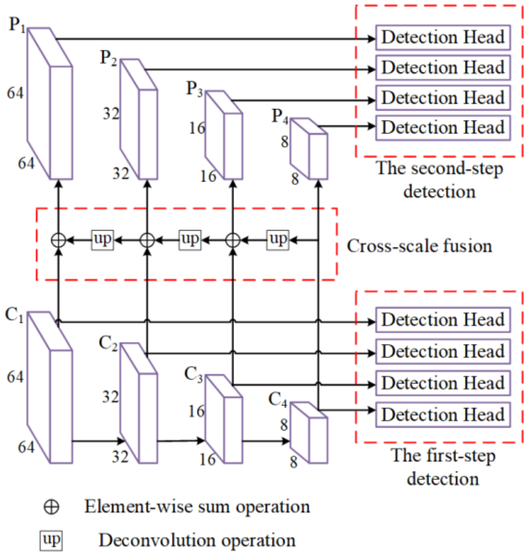


Fig. 5. The architecture of the detection module.

2) Feature extraction network: We use ResNet50 [9] as the feature extraction network (FEN). ResNet50 significantly increases the number of network layers through the design of skip connection, and avoids the problems of gradient disappearance and gradient explosion. And the ResNet50 is extensively utilized in SAR ship detection due to its powerful feature extraction capability.

3) Detection branch: The detection branch (DB) takes the output of the FEN, i.e., the F_f, as input. And the DB can be divided into the detection module and the detection loss, as shown in Fig. 5. The detection module is the most important module of RefineDet, which performs cross-scale fusion and two-step detection operations. And we did not modified it during use. Specifically, the detection module first extracts multi-scale features $\mathbf{C}_\mathbf{k}$, where $\mathrm{k} \in [1,2,3,4]$ represents the k-th scale feature map. Then, the cross-scale fusion is performed on $\mathbf{C}_\mathbf{k}$ to obtain $\mathbf{P}_\mathbf{k}$. This operation makes the feature information in $\mathbf{P}_\mathbf{k}$ more abundant and beneficial for detecting small ships. During the process of two-step detection, the predefined anchor boxes and $\mathbf{C}_\mathbf{k}$ are inputted into the detection head for the initial step of detection to generate rough bounding boxes b'. Then in the second step of detection, the b' and $\mathbf{P}_{\mathbf{k}}$ are inputted into the detection head to generate fine bounding boxes b. The two-step detection strategy is beneficial to filter out the majority of false alarms and enhance the localization accuracy of the bounding boxes.

The definition of detection loss is shown in (1):

$$L_{\mathrm{DET}}=\frac{1}{N_1}\left[\sum_{\mathrm{b}} L_{\mathrm{cls}}(p_{\mathrm{b}1},p_{\mathrm{b}}^*)+\sum_{\mathrm{b}} p_{\mathrm{b}}^* L_{\mathrm{reg}}(t_{b1},t_b^*)\right] +\frac{1}{N_2}\left[\sum_{\mathrm{b}} L_{\mathrm{cls}}(p_{\mathrm{b}2},p_{\mathrm{b}}^*)+\sum_{\mathrm{b}} p_{\mathrm{b}}^* L_{\mathrm{reg}}(t_{b2},t_b^*)\right]. \quad (1)$$

where L_{cls} and L_{reg} represent the cross-entropy loss and smooth L1 loss, respectively. N_1 denotes the number of preset anchor boxes, $p_{\mathrm{b}1}$ and t_{b1} denote the confidence score and the regression target, respectively. p_{b}^* and t_b^* denote the class label and the location label, respectively. The definition of N_2, $p_{\mathrm{b}2}$, and t_{b2} in the second step detection exhibits similarities to those of N_1, $p_{\mathrm{b}1}$, and t_{b1}. The details of the DB can also be traced in [2].

4) SCMB: The SCMB is the core part of the SCMS-Net. Same with the DB, it also takes the output of the FEN as input. The SCMB is designed to utilize the scene-level labels and assist the FEN in learning the characteristics of different scenes. We concretely introduce the details of SCMB in Section II.B.

B. Scene Characteristic Mining Branch

From Fig. 2, it is evident that different scenes display distinct characteristics. Ships may exist in both inshore and offshore SAR images. However, the inshore clutter is mainly harbor facilities and containers, and the offshore clutter is mainly sea clutter and reef. For the inland SAR images, there are no ship targets, and the clutter is mainly buildings, roads, and trees. This leads us to naturally think that learning these characteristics of different scenes is essentially acquiring effective feature representation for ships and clutter, which greatly benefits SAR ship detection. Therefore, we designed the SCMB, as shown in Fig. 6. It can help the network learn the characteristics of different scenes by using scene-level labels. The SCMB consists of the scene classification module (SCM), the scene classification loss (SCL), the scene aggregation module (SAM), and the scene aggregation loss (SAL). Next, we will introduce them in detail.

1) SCM and SCL: The SCM is responsible for recognizing the scene category of the SAR image. It takes the output of the FEN as input. Then, the F_f passes through a max-pooling layer and two fully connected (FC) layers sequentially. Subsequently, we employ a softmax layer to generate scene classification result s.

The scene classification loss is computed using the s and the corresponding scene-level labels s^*:

$$L_{\mathrm{SC}}=-\frac{1}{N}\sum_{\mathrm{i}=1}^{N}\left[s_{\mathrm{i}}\log(s_{\mathrm{i}}^*)+(1-s_{\mathrm{i}})\log(1-s_{\mathrm{i}}^*)\right]. \quad (2)$$

By imposing the constraint of the SCL, the SCM can accurately classify the scene category depicted in SAR images, while the FEN learns to extract discriminative features that differentiate different scenes. In this manner, the FEN can effectively learn the characteristics of different scenes.

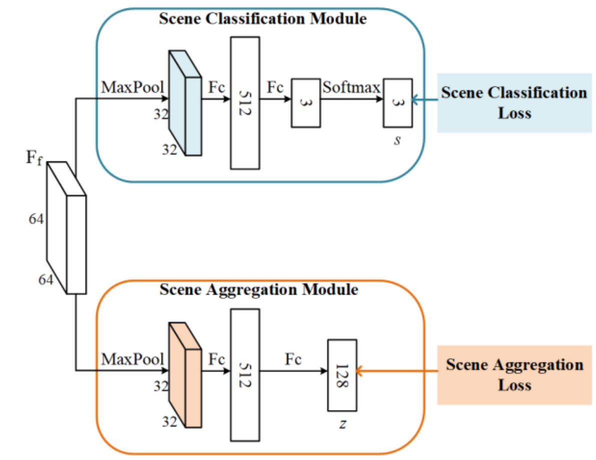


Fig. 6. The structure of the SCMB.

2) SAM and SAL: Same with the SCM, the SAM also takes the output of the FEN as input. But different from SCM, under the constraints of SAM, the SAM is used to generate embedded features that can embody intra-class consistency. In the SAM, the F_f first passes through a max-pooling layer and two FC layers sequentially. Then, we can obtain the embedded feature of SAR image: z.

The SAL is computed using the embedded feature z and the corresponding scene-level labels:

$$L_{\mathrm{SA}}=-\frac{1}{N}\sum_{\mathrm{i}=1}^{N}\sum_{\mathrm{u}=1}^{N} I_{[u\neq \mathrm{i}]} I_{[s_u^*=s_{\mathrm{i}}^*]} \log\frac{\exp(\frac{c(z_{\mathrm{i}},z_u)}{\tau})}{\sum_{v=1}^{N} I_{[v\neq \mathrm{i}]}\exp(\frac{c(z_{\mathrm{i}},z_v)}{\tau})}. \quad (3)$$

where τ denotes a scalar temperature parameter, $c(g,h)$ denotes the cosine similarity between l_2 normalized g and h:

$$c(g,h)=\frac{g^T h}{\|g\|\|h\|}. \quad (4)$$

$I_{[u\neq \mathrm{i}]}$, $I_{[sr_u^*=sr_{\mathrm{i}}^*]}$, and $I_{[u\neq \mathrm{i}]}$ are indicator functions:

$$I_{[u\neq \mathrm{i}]}=\begin{cases}1 & \text{if } \mathrm{u}\neq \mathrm{i}\\ 0 & \text{otherwise}\end{cases},\ I_{[s_u^*=s_{\mathrm{i}}^*]}=\begin{cases}1 & \text{if } s_u^*=s_{\mathrm{i}}^*\\ 0 & \text{otherwise}\end{cases}. \quad (5)$$

$$I_{[v\neq \mathrm{i}]}=\begin{cases}1 & \text{if } \mathrm{v}\neq \mathrm{i}\\ 0 & \text{otherwise}\end{cases}. \quad (6)$$

By imposing the constraint of the SAL, the feature similarity among SAR images within each scene is enlarged, while the FEN can learn to extract consistent features from the same scene. This enables the FEN to better capture the characteristics of each scene.

III. TRAINING AND TEST PROCESS OF THE PROPOSED METHOD

A. Semi-supervised Training Process

During the semi-supervised training process, the entire network is optimized jointly using the two types of labeled SAR images. The overall loss function is formulated as:

$$L=\lambda_1 L_{\mathrm{SC}}+\lambda_2 L_{\mathrm{SA}}+L_{\mathrm{DET}}. \quad (7)$$

where λ_1 and λ_2 are the weight factors. They are used to balance the L_{SC}, L_{SA}, and L_{DET}.

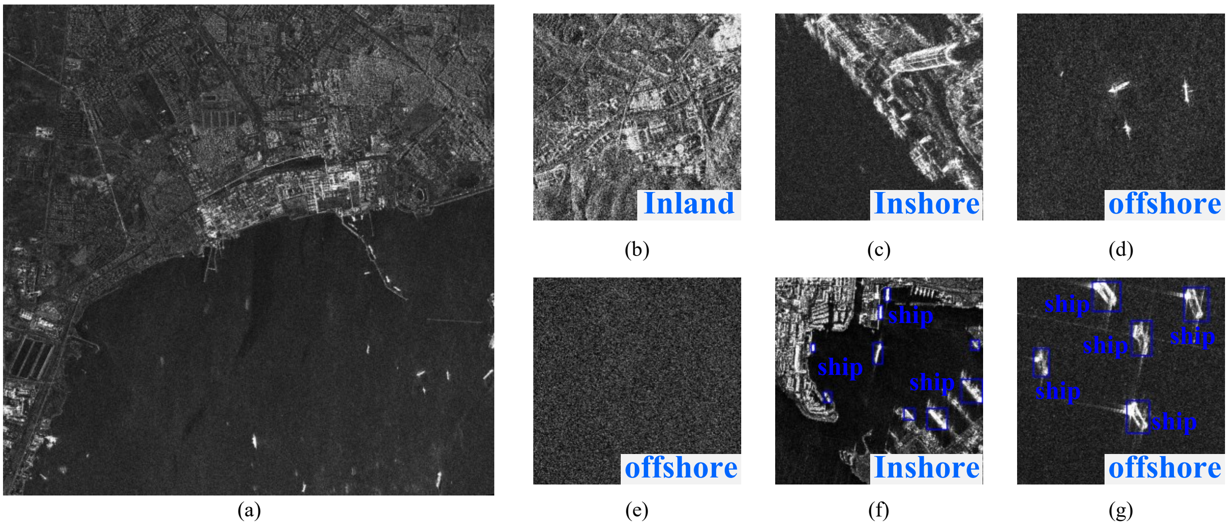


Fig. 7. Examples from AIR-SARShip-1.0 dataset. (a) an original large-scale SAR image; (b-e) some sub-images with only scene-level labels; (f-g) some sub-images with both scene-level and target-level labels.

Algorithm 1. The Hierarchical Test Process

Input: test SAR image x, the trained network $net(\cdot)$, the confidence thresholds th_{ins} and th_{offs};

Output: the detection result dr;

1. $F_f \leftarrow net_{FEM}(x)$;
2. $s \leftarrow net_{SCM}(F_f)$;
3. **if** $s ==$ 'inland scene' **then**
4. $\quad dr = \varnothing$;
5. **else if** $s ==$ 'offshore scene' **then**
6. $\quad b \leftarrow net_{DM}(F_f)$;
7. $\quad dr \leftarrow f_{nms}(f_{filter}(b, th_{offs}))$;
8. **else then**
9. $\quad b \leftarrow net_{DM}(F_f)$
10. $\quad dr \leftarrow f_{nms}(f_{filter}(b, th_{ins}))$
11. **end if**
12. **return** dr

B. Hierarchical Test Process

The hierarchical test process is outlined in Algorithm 1, which consists of two main steps: scene classification and ship detection. During the scene classification step, the test SAR image is initially fed into the FEN to generate F_f. Subsequently, the F_f is fed into the SCM to generate the scene classification result s. In the ship detection step, we employ specialized detection strategies that are specifically designed for each scene. For inland scenes, the detection results are empty because ships can not appear in inland areas. For inshore scenes, the F_f will be fed into the detection module to generate the predicted bounding boxes. And then, we apply the threshold th_{ins} to filter out the bounding boxes with low confidence. For offshore scenes, after the detection module generates the predicted bounding boxes, we apply the threshold th_{offs} to filter them. Finally, the non-maximum suppression is performed to generate final detection results. The traditional test process generally set a single confidence threshold for all SAR images, regardless of their scene types. However, the clutter complexity varies between inshore and offshore scenes, thus the optimal confidence threshold should be different. Intuitively, th_{ins} should be set higher than th_{offs} because the clutter in inshore scenes is more complex, which increases the likelihood of false alarms. In our experiments, we determine the th_{ins} and th_{offs} through cross-validation.

In the design of the hierarchical test process, we have taken into full consideration the characteristics of different scenes. This test process can facilitate the reduction of false alarms in inland and inshore scenes, while also reducing missing ships in offshore scenes, thereby achieving higher detection performance.

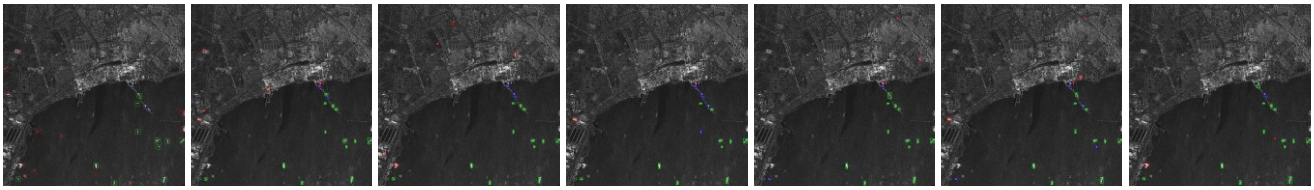
IV. EXPERIMENTS

A. Dataset

In our experiments, we choose the measured AIR-SARShip-1.0 dataset [10] to evaluate the SCMS-Net. This dataset contains a total of 31 SAR images. And each image has a size of 3000×3000 pixels. For our experiments, we used 21 images for training and validation, while the remaining 10 images were reserved for testing. During training, due to the limited GPU memory, we cropped the original training images to obtain multiple fixed-size subimages. After considering the ship size and GPU memory comprehensively, we set the size of subimage to 512×512. All training subimages were labeled with scene-level labels, regardless of the presence of ships. And then we randomly selected 30% of the training sub-images containing ships to label the target-level labels. During the test process, we use a sliding window with a size of 512 to crop the large-scale test images into multiple subimages. Finally, we restore the detection results on all test subimages to the original large-scale images. Fig. 7 illustrates some examples of SAR images from our dataset.

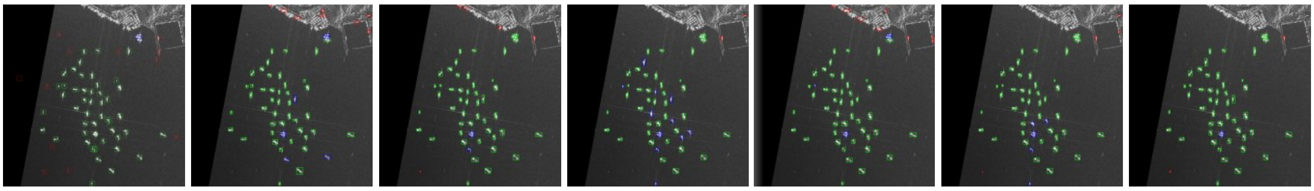
B. Implementation details

The stochastic gradient descent (SGD) optimizer is used to optimize the entire network, and the learning rate is set to 0.001. Two training hyperparameters are introduced in the proposed network, in which the λ_1 and λ_2 in (7) are both set to 0.1.



(a) (b) (c) (d) (e) (f) (g)

Fig. 8. The detection results on the large-scale test SAR image I. (a) Faster R-CNN. (b) RefineDet. (c) YOLOX. (d) STAC. (e) Soft Teacher. (f) SCMS-Net.



(a) (b) (c) (d) (e) (f) (g)

Fig. 9. The detection results on the large-scale test SAR image II. (a) Gaussian-CFAR. (b) Faster R-CNN. (c) RefineDet. (d) YOLOX. (e) STAC. (f) Soft Teacher. (g) SCMS-Net.

C. *Evaluation Metrics*

We choose the False alarm ratio (FA), Precision, Recall, and F1-score metrics, which are commonly used in SAR target detection as the evaluation metrics:

$$\text{FA} = \frac{\text{the number of incorrectly detected ships}}{\text{the number of detected ships}}. \quad (8)$$

$$\text{Precision} = \frac{\text{the number of correctly detected ships}}{\text{the number of detected ships}}. \quad (9)$$

$$\text{Recall} = \frac{\text{the number of correctly detected ships}}{\text{the ground truth number of ships}}. \quad (10)$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (11)$$

D. *Comparison with Other Methods*

In this section, we contrast the SCMS-Net with six other powerful detectors, including a traditional unsupervised method, i.e., Gaussian-CFAR [1], three fully supervised methods, i.e., the Faster R-CNN [5], RefineDet [3], and YOLOX [4], and two semi-supervised methods, i.e., the STAC [8] and Soft Teacher [7]. It is worth noting that the training subimages with target-level labels make up 30% of all training subimages containing ships in our experiments. Fig. 8 and Fig. 9 show the visualized detection results on two large-scale test SAR images. The green boxes, red boxes, and blue boxes represent the correctly detected ships, false alarms, and missing ships, respectively. Table I provides the quantitative experimental results of different methods.

As observed in Fig. 8 and Fig. 9, the Gaussian-CFAR method has the worst detection performance, especially it produces lots of false alarms. The three fully supervised detectors also generated many false alarms and missing ships. Comparatively, the two semi-supervised detectors, i.e., the STAC and Soft Teacher, achieved relatively better detection results due to their utilization of unlabeled subimages. However, in some complex inland and inshore scenes, both of these two methods still exhibit a high false alarm rate. The SCMS-Net outperforms other methods, as evident in Fig.8 (f) and Fig. 9 (f). The SCMS-Net generates only a very limited number of false alarms and missing ships. As shown in Table I, the SCMS-Net outperforms other comparative methods. This is mainly attributed to the utilization of scene-level labels

TABLE I. EXPERIMENTAL RESULTS OF DIFFERENT METHODS

Method	FA (%)	Precision (%)	Recall (%)	F1-score (%)	Total Test Time (seconds)
Gaussian-CFAR	49.3	50.7	85.6	63.69	142.42
Faster R-CNN	26.57	73.43	84.80	78.70	140.51
RefineDet	27.03	72.97	87.20	79.45	21.16
YOLOX	18.49	81.51	78.40	79.93	23.76
STAC	23.94	76.06	85.60	80.55	140.51
Soft Teacher	21.54	78.46	84.00	81.14	140.51
SCMS-Net	**16.28**	**83.72**	**88.80**	**86.18**	**16.63**

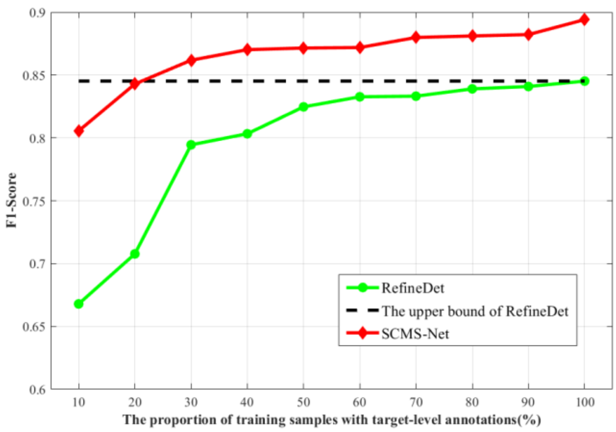


Fig.10. The performance of SCMS-Net and RefineDet under different proportions of training subimages with target-level labels.

and the adoption of a more flexible and targeted test process. Furthermore, the SCMS-Net has the shortest total test time. This is because the inland SAR images did not go through the detection module to generate detection results, thereby improving the test efficiency. In conclusion, the SCMS-Net achieves significant improvements in both detection performance and test efficiency.

As the proportion of training samples with target-level labels increased from 10% to 100% among the training samples containing ships, we analyzed the performance variations of the SCMS-Net and our baseline, i.e., the

RefineDet. Fig. 10 shows their experimental results. As observed in Fig. 10, the SCMS-Net consistently exhibits better detection performance across all proportion values compared to RefineDet. Especially when the proportion values are 10% and 20%, the performance of the SCMS-Net is much higher than that of RefineDet. And when the proportion value reaches 30%, the SCMS-Net surpasses the upper bound of RefineDet. It demonstrates that the SCMS-Net can achieve higher performance when target-level labels are limited, by effectively leveraging the scene-level labels.

V. CONCLUSION

A novel SCMS-Net is proposed in this paper. In the SCMS-Net, a well-designed SCMB can assist in improving detection performance when target-level labels are limited, by effectively leveraging the scene-level labels. And a more flexible and targeted hierarchical test process is designed to reduce the false alarms in inland and inshore scenes, as well as missing ships in the offshore scene. The experiments based on an authoritative measured dataset validate the effectiveness of the SCMS-Net.

ACKNOWLEDGMENT

This work was supported in part by the National Science Foundation of China under Grant U21B2039, in part by the 111 Project.

REFERENCES

[1] L. M. Novak, M. C. Burl, and W. W. Irving, "Optimal polarimetric processing for enhanced target detection," IEEE Trans. Aerosp. Electron. Syst, vol. 29, no. 1, pp. 234–244, Jan. 1993.

[2] Y. Du, L. Du, and L. Li, "An SAR Target Detector Based on Gradient Harmonized Mechanism and Attention Mechanism," IEEE Geoscience and Remote Sensing Letters, vol. 19, pp. 1-5, 2022.

[3] S. Zhang, L. Wen, X. Bian, Z. Lei, and S. Z. Li, "Single-Shot Refinement Neural Network for Object Detection," in Proc. IEEE Comput. Vis. Pattern Recognit. (CVPR), 2018, pp. 4203-4212.

[4] G. Zheng, S. Liu, F. Wang, Z. Li, and J. Sun, "Yolox: Exceeding yolo series in 2021," 2021 arXiv: 2107.08430.

[5] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in Proc. Conf. Neural Inf. Process. Syst. (NeurIPS), 2015, pp. 91–99.M. Young, The Technical Writer's Handbook. Mill Valley, CA: University Science, 1989.

[6] Q. An, Z. Pan, L. Liu, and H. You, "DRBox-v2: An Improved Detector With Rotatable Boxes for Target Detection in SAR Images, " in IEEE Transactions on Geoscience and Remote Sensing, vol. 57, no. 11, pp. 8333-8349, Nov. 2019.

[7] M. Xu, Z. Zhang, H. Hu, J. Wang, L. Wang, F. Wei, X. Bai, and Z. Liu, "End-to-End Semi-Supervised Object Detection with Soft Teacher," in IEEE/CVF International Conference on Computer Vision (ICCV), 2021, pp. 3040-3049.

[8] K. Sohn, Z. Zhang, C.-L. Li, H. Zhang, C.-Y. Lee, and T. Pfister, "A simple semi-supervised learning framework for object detection," 2020, arXiv:2005.04757.

[9] K. He, X. Zhang, S. Ren, et al. "Deep residual learning for image recognition," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 770-778.

[10] X. Sun, Z. Wang, Y. Sun, W. Diao, Y. Zhang, and K. Fu, "AIRSARShip-1.0: High-resolution SAR ship detection dataset," J. Radars, vol. 8, pp. 852–862, Dec. 2019.